

CAL. LIC. # 434901

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*AIR AND HYDRONIC SYSTEM
TESTING AND BALANCING*



TEST AND BALANCE SURVEY

NANOTECH ENERGY INC FUME HOOD SURVEY 2024

MECHANICAL CONTRACTOR: _____

CONSULTING ENGINEER: _____

JOB NO: _____ 20531C

DATE: _____ DECEMBER 13, 2024

TECHNICIAN: _____ DANIEL PALLAS &
MICHAEL CONNOLLY

TBE: _____ MATTHEW I. BRENNAN



NANOTECH ENERGY INC FUME HOOD SURVEY 2024

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GENERAL NOTES

1. CORRECTIONS FOR TEMPERATURE AND ALTITUDE HAVE BEEN MADE ON ALL TEST RESULTS SHOWN IN THIS REPORT.
2. BALANCE FACTORS FOR EXHAUST SIDEWALL AND SUPPLY GRILLES WERE CALCULATED FROM THE CORE AREA AND MEASURED WITH A CALIBRATED METER, MODEL, M.F.G. (SEE INSTRUMENT CERT. OF CALIBRATION.)
3. ALL INSTRUMENTS USED FOR TESTING AND BALANCING ARE TRACEABLE BACK TO THE NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY AND ARE TESTED IN OUR OWN LAB PER AABC.
4. FOLLOWING THIS SHEET ARE:
 - B. SYMBOL SHEET
 - C. INSTRUMENT SHEETS

ACRONYM SHEET

ACH	AIR CHANGES PER HOUR	HD	HEAD
ADD	ADDRESS	HD	HOT DECK
ADJ. P.D.	ADJUSTED PITCH DIAMETER	HP	HORSEPOWER
BHP	BRAKE HORSEPOWER	IC	INCHES OF WATER COLUMN
CL	CENTER	KW	KILOWATTS
CD	CEILING DIFFUSER	LA	LEAVING AIR TEMPERATURE
CD	COLD DECK	LWT	LEAVING WATER TEMPERATURE
CFM	CUBIC FEET PER MINUTE	MBH	THOUSAND BTU PER HOUR
COMP	COMPRESSOR	MIN	MINIMUM
CONT	CONTROLLER	N/A	NOT ACCESSIBLE / NOT AVAILABLE
CORR	CORRECTION	N/T	NOT TAKEN DUE TO NO VALID LOCATION
CR	CEILING RETURN	OA	OUTSIDE AIR
DB	DRY BULB TEMPERATURE	OPEN	NO TERMINAL DEVICE INSTALLED
DD	DIRECT DRIVE	PD	PITCH DIAMETER
DIA	DIAMETER	PP	PERFORATED PLATE
DNA	DATA NOT AVAILABLE	PSI	POUNDS PER SQUARE INCH
DNL	DATA NOT LISTED	RPM	REVOLUTIONS PER MINUTE
ΔP OR DP	PRESSURE DIFFERENTIAL	SCP	STATIC CONTROL POINT
(E)	EXISTING	SF	SERVICE FACTOR
EA	ENTERING AIR TEMPERATURE	SP	STATIC PRESSURE
EA	EFFECTIVE AREA / K FACTOR	SP	SETPOINT
ESP	EXTERNAL STATIC PRESSURE	SWE	SIDE WALL EXHAUST
EWI	ENTERING WATER TEMPERATURE	SWR	SIDE WALL RETURN
FE	FLOOR EXHAUST	SWS	SIDE WALL SUPPLY
FLA	FULL LOAD AMPS	TEMP	TEMPERATURE
FPM	FEET PER MINUTE VELOCITY	TSP	TOTAL STATIC PRESSURE
FR	FLOOR RETURN	V	VOLTS
FS	FLOOR SUPPLY	VP	VELOCITY
FT	FEET	WB	WET BULB TEMPERATURE
GDR	GRILLE, DIFFUSER, REGISTER	#	NUMBER OR POUND
GPM	GALLONS PER MINUTE		

Test & Balance Instrumentation Sheet

- ☒ THE FOLLOWING CHECKED INSTRUMENTS WERE USED TO SUCCESSFULLY MEASURE AND SET EACH DEVICE ON THIS PROJECT.

	<u>Manufacturer</u>	<u>Model No.</u>	<u>Serial No.</u>	<u>Calibration Date</u>
☐ AIR DATA MULTIMETER	SHORTRIDGE	ADM-860C	M20253	3/1/2024
☐ FLOW HOOD	SHORTRIDGE	DIGITAL 8400		N/A
☒ FLOW MODULE	EVERGREEN	S-PVF-1	24D-00460	6/11/2024
☐ CAPTURE HOOD	EVERGREEN	CH-15D	2200137	N/A
☐ ROTATING VANE ANEMOMETER	ALNOR	RVA801	A05381	3/1/2024
☐ AMPROBE	FLUKE	325	40020345WS	3/1/2024
☐ TACHOMETER	MITUTOYO	PH-100-A	A16A95019P	3/1/2024
☐ HYDRONIC MANOMETER	TSI	HM-685	72009029	2/27/2024
☐ SOUND METER	QUEST TECH.	2900 SLM	CDC060026	5/1/2024
☐ VIBRATION	BALMAC INC.	200	0811096	3/26/2024
☐ POUND GAUGE	DWYER DIGITAL	DPG-005	4366516	3/1/2024
☐ HUMIDITY & TEMPERATURE	FLUKE #2	971	38401145	3/1/2024
☐ ULTRASONIC WATER FLOW METER	FUJI ELECTRIC	FSCS	N0F0844T	2/27/2024

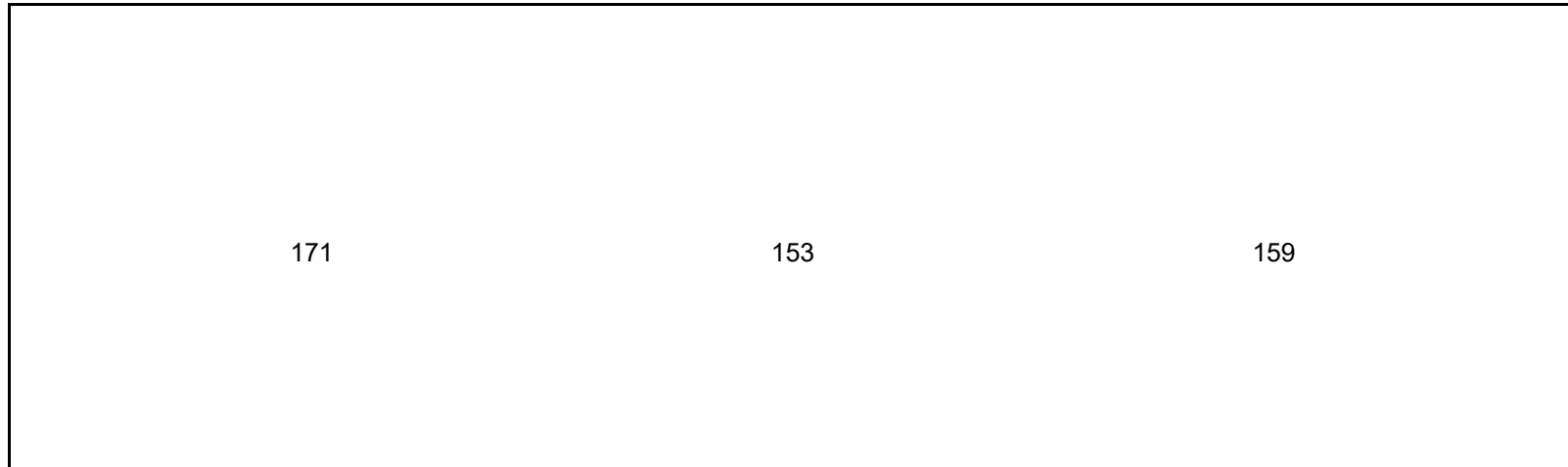


LABORATORY HOOD TEST

Job Number: 20531C
Hood #: 1

Date: 11/26/2024
Room #: LAB

RECORDED VELOCITY READING



SIZE OF HOOD OPENING	12.5 "	x	39 "	=	3.39	SQ.FT.
LOWEST VELOCITY	153					
AVERAGE VELOCITY	161	x	3.39	SQ.FT. =	545	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	6/11/2024	
TECHNICIAN	DANIEL PALLAS	TBE		MATT BRENNAN		
SASH MARKER INSTALLED	12.5					
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks:



LABORATORY HOOD TEST

Job Number: 20531C
Hood #: 2

Date: 11/26/2024
Room #: LAB

RECORDED VELOCITY READING

120	130	118
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SIZE OF HOOD OPENING	12.5 "	x	39 "	=	3.39	SQ.FT.
LOWEST VELOCITY	118					
AVERAGE VELOCITY	123	x	3.39	SQ.FT. =	415	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	6/11/2024	
TECHNICIAN	DANIEL PALLAS	TBE		MATT BRENNAN		
SASH MARKER INSTALLED	12.5					
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks:

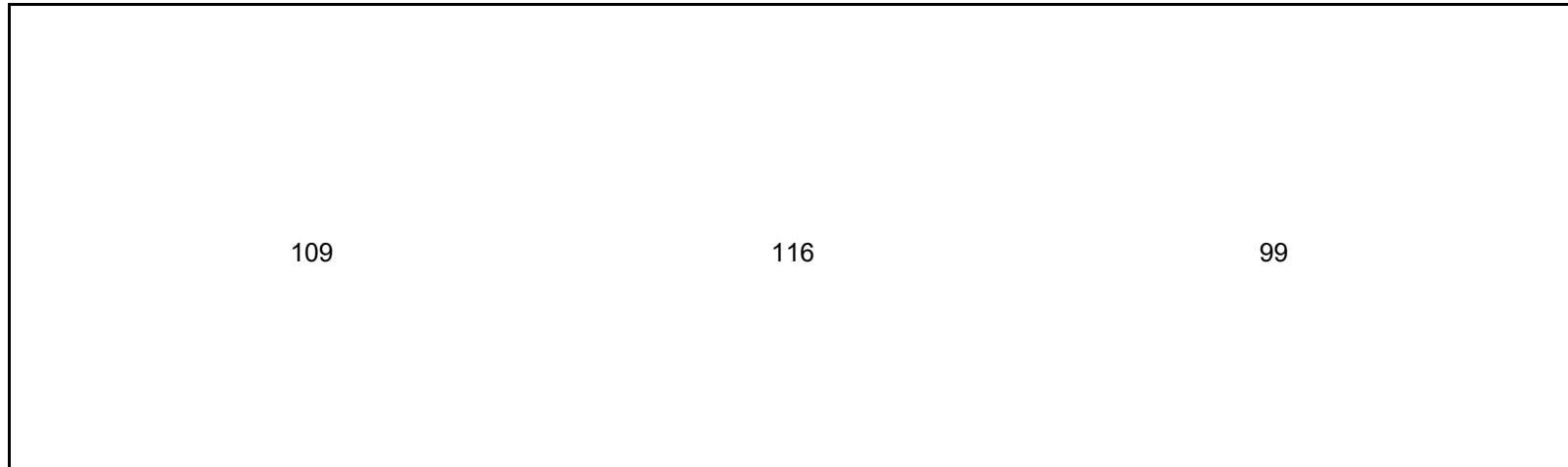


LABORATORY HOOD TEST

Job Number: 20531C
Hood #: 3

Date: 11/26/2024
Room #: LAB

RECORDED VELOCITY READING



SIZE OF HOOD OPENING	12.5 "	x	39 "	=	3.39	SQ.FT.
LOWEST VELOCITY	99					
AVERAGE VELOCITY	108	x	3.39	SQ.FT. =	366	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	6/11/2024	
TECHNICIAN	DANIEL PALLAS	TBE		MATT BRENNAN		
SASH MARKER INSTALLED	12.5					
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks:



LABORATORY HOOD TEST

Job Number: 20531C
Hood #: 4

Date: 11/26/2024
Room #: LAB

RECORDED VELOCITY READING

140	108	115
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SIZE OF HOOD OPENING	12.5 "	x	39 "	=	3.39	SQ.FT.
LOWEST VELOCITY	108					
AVERAGE VELOCITY	121	x	3.39	SQ.FT. =	410	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	6/11/2024	
TECHNICIAN	DANIEL PALLAS	TBE		MATT BRENNAN		
SASH MARKER INSTALLED	12.5					
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks:

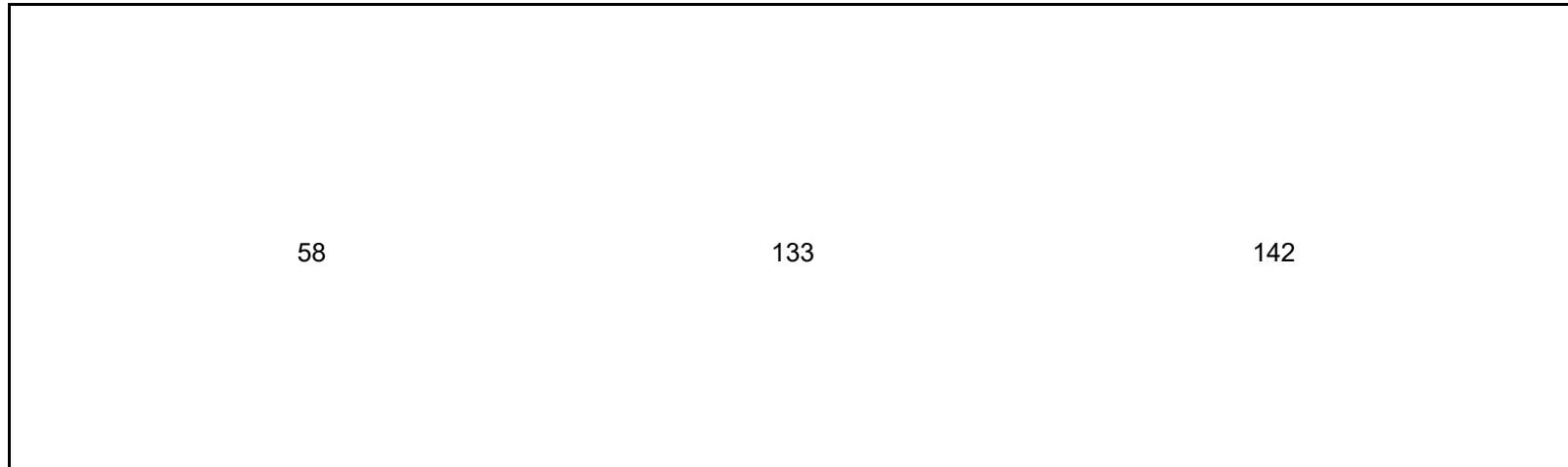


LABORATORY HOOD TEST

Job Number: 20531C
Hood #: 5

Date: 11/26/2024
Room #: LAB

RECORDED VELOCITY READING



SIZE OF HOOD OPENING	12.5 "	x	39 "	=	3.39	SQ.FT.
LOWEST VELOCITY	58					
AVERAGE VELOCITY	111	x	3.39	SQ.FT. =	376	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	6/11/2024	
TECHNICIAN	DANIEL PALLAS	TBE		MATT BRENNAN		
SASH MARKER INSTALLED	12.5					
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks:



LABORATORY HOOD TEST

Job Number: 20531C
Hood #: 6

Date: 11/26/2024
Room #: LAB

RECORDED VELOCITY READING

138	135	123
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SIZE OF HOOD OPENING	12.5 "	x	39 "	=	3.39	SQ.FT.
LOWEST VELOCITY	123					
AVERAGE VELOCITY	132	x	3.39	SQ.FT. =	447	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	6/11/2024	
TECHNICIAN	DANIEL PALLAS	TBE		MATT BRENNAN		
SASH MARKER INSTALLED	12.5					
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks:



LABORATORY HOOD TEST

Job Number: 20531C
Hood #: 7

Date: 11/26/2024
Room #: LAB

RECORDED VELOCITY READING

117	112	105	100	93	71
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SIZE OF HOOD OPENING	12.5 "	x	87 "	=	7.55	SQ.FT.
LOWEST VELOCITY	71					
AVERAGE VELOCITY	100	x	7.55	SQ.FT. =	753	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	6/11/2024	
TECHNICIAN	DANIEL PALLAS	TBE		MATT BRENNAN		
SASH MARKER INSTALLED	12.5					
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks:



LABORATORY HOOD TEST

Job Number: 20531C
Hood #: 8

Date: 11/26/2024
Room #: LAB

RECORDED VELOCITY READING

102	92	93	100	114	117
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SIZE OF HOOD OPENING	12.5 "	x	87 "	=	7.55	SQ.FT.
LOWEST VELOCITY	92					
AVERAGE VELOCITY	103	x	7.55	SQ.FT. =	778	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	6/11/2024	
TECHNICIAN	DANIEL PALLAS	TBE		MATT BRENNAN		
SASH MARKER INSTALLED	12.5					
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks:



LABORATORY HOOD TEST

Job Number: 20531C
Hood #: 9

Date: 11/26/2024
Room #: LAB

RECORDED VELOCITY READING

94	92	97	97	91	94
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SIZE OF HOOD OPENING	12.5 "	x	87 "	=	7.55	SQ.FT.
LOWEST VELOCITY	91					
AVERAGE VELOCITY	94	x	7.55	SQ.FT. =	711	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	6/11/2024	
TECHNICIAN	DANIEL PALLAS	TBE		MATT BRENNAN		
SASH MARKER INSTALLED	12.5					
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks:

DIFFUSER AND GRILLE TEST SHEET

Job Number: 20531C

Date: 12/12/2024
System: Canopy Hoods

[illegible]Remarks:



LABORATORY HOOD TEST

Job Number: 20531C
Hood #: 1

Date: 12/12/2024
Room #: Graphene

RECORDED VELOCITY READING

53	65	65	64	56
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SIZE OF HOOD OPENING	67 "	x	18 "	=	8.38	SQ.FT.
LOWEST VELOCITY	53					
AVERAGE VELOCITY	61	x	8.38	SQ.FT. =	508	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	4/18/2024	
TECHNICIAN	MICHAEL CONNOLLY	TBE		MATT BRENNAN		
SASH MARKER INSTALLED	YES					
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks:



LABORATORY HOOD TEST

Job Number: 20531C
Hood #: 1

Date: 12/12/2024
Room #: EL. LAB

RECORDED VELOCITY READING

102	91	91	92	89	92	106
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SIZE OF HOOD OPENING	85 "	x	12 "	=	7.08	SQ.FT.
LOWEST VELOCITY	89					
AVERAGE VELOCITY	95	x	7.08	SQ.FT. =	671	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	4/18/2024	
TECHNICIAN	MICHAEL CONNOLLY	TBE		MATT BRENNAN		
SASH MARKER INSTALLED	YES					
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks:



LABORATORY HOOD TEST

Job Number: 20531C
Hood #: 2

Date: 12/12/2024
Room #: EL. LAB

RECORDED VELOCITY READING

114	102	105	95	85
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SIZE OF HOOD OPENING	62 "	x	12 "	=	5.17	SQ.FT.
LOWEST VELOCITY	85					
AVERAGE VELOCITY	100	x	5.17	SQ.FT. =	518	CFM EXHAUST
INSTRUMENT	S-PVF-1			DATE CALIBRATED	4/18/2024	
TECHNICIAN	MICHAEL CONNOLLY	TBE		MATT BRENNAN		
SASH MARKER INSTALLED	YES					
READINGS TAKEN ON 14" CENTERS USING 16 POINT AVERAGE						

Remarks: